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$$\begin{aligned} & \int \frac{\sin x + x \cos x}{x \sin x} dx \\ &= \int \frac{\sin x}{x \sin x} dx + \int \frac{x \cos x}{x \sin x} dx \\ &= \int \frac{1}{x} dx + \int \frac{\cos x}{\sin x} dx \\ &= \ln |x| + \int \frac{\cos x}{\sin x} dx \end{aligned}$$

Substitutie: $t = \sin x \Rightarrow dt = \cos x dx$

$$\begin{aligned} &= \ln |x| + \int \frac{1}{t} dt \\ &= \ln |x| + \ln |t| + c \\ &= \ln |x| + \ln |\sin x| + c \end{aligned}$$

References